

Integration by Method of Substitution:

Integration by method of substitution involves the introduction of a function that changes the integrand into an easy integral form.

Note:

This method is commonly used for irrational functions. Some useful substitutions are given in table below

Expression Involving	Suitable Substitution
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$
$\sqrt{x \pm a}$	$\sqrt{x \pm a} = t$
$\sqrt{2ax - x^2}$	$x - a = a \sin \theta$
$\sqrt{2ax + x^2}$	$x + a = a \sec \theta$

Some Important Results:

$$\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + c$$

$$\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + c$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + c$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \left(\frac{x}{a} \right) + c$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left| x + \sqrt{x^2 - a^2} \right| + c$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left| x + \sqrt{x^2 + a^2} \right| + c$$

$$\int \frac{1}{|x| \sqrt{x^2 - a^2}} dx = \frac{1}{a} \sec^{-1} \left(\frac{x}{a} \right) + c$$

$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + c$$

$$\int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \ln \left| x + \sqrt{x^2 - a^2} \right| + c$$

$$\int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \ln \left| x + \sqrt{x^2 + a^2} \right| + c$$

EXERCISE 3.3**Evaluate the following integrals**

Q.1 $\int \frac{-2x}{\sqrt{4-x^2}} dx$

Solution:

$$\begin{aligned} & \int \frac{-2x}{\sqrt{4-x^2}} dx \\ &= \frac{(4-x^2)^{\frac{1}{2}+1}}{\left(-\frac{1}{2}+1\right)} + c \\ &= \frac{(4-x^2)^{\frac{1}{2}}}{\left(\frac{1}{2}\right)} + c \\ &= 2\sqrt{4-x^2} + c \end{aligned}$$

Q.2 $\int \frac{dx}{x^2+4x+13}$

Solution:

$$\begin{aligned} & \int \frac{dx}{x^2+4x+13} \\ &= \int \frac{dx}{(x^2+4x+4)+9} \\ &= \int \frac{dx}{(x+2)^2+(3)^2} \end{aligned}$$

$$\begin{aligned} \text{Using } \int \frac{1}{a^2+x^2} dx &= \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + c \\ &= \frac{1}{3} \tan^{-1}\left(\frac{x+2}{3}\right) + c \end{aligned}$$

Q.3 $\int \frac{x^2}{4+x^2} dx$

Solution:

$$\begin{aligned} & \frac{x^2}{4+x^2} \text{ is an improper fraction} \\ & \text{By long division} \\ & x^2 + 4 \overline{\begin{array}{r} 1 \\ x^2 \\ \hline \pm x^2 \pm 4 \\ \hline -4 \end{array}} \end{aligned}$$

$$\begin{aligned} \therefore \int \frac{x^2}{4+x^2} dx &= \int \left(1 - \frac{4}{4+x^2}\right) dx \\ &= \int 1 dx - 4 \int \frac{1}{4+x^2} dx \\ &= x - 4 \int \frac{1}{2^2+x^2} dx \end{aligned}$$

$$\begin{aligned} \text{Using } \int \frac{1}{a^2+x^2} dx &= \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + c \\ &= x - 4 \left(\frac{1}{2} \tan^{-1} \frac{x}{2} \right) + c \\ &= x - 2 \tan^{-1} \frac{x}{2} + c \end{aligned}$$

Q.4 $\int \frac{1}{x \ln x} dx$

Solution:

$$\begin{aligned} & \int \frac{1}{x \ln x} dx \\ &= \int \frac{\left(\frac{1}{x}\right)}{\ln x} dx \\ &= \ln |\ln x| + c \end{aligned}$$

Q.5 $\int \frac{e^x}{e^x+3} dx$

Solution:

$$\begin{aligned} & \int \frac{e^x}{e^x+3} dx \\ &= \ln |e^x+3| + c \end{aligned}$$

Q.6 $\int \frac{x+b}{(x^2+2bx+c)^{\frac{1}{2}}} dx$

Solution:

$$\begin{aligned} & \int \frac{x+b}{(x^2+2bx+c)^{\frac{1}{2}}} dx \\ &= \frac{1}{2} \int \frac{2x+2b}{(x^2+2bx+c)^{\frac{1}{2}}} dx \\ &= \frac{1}{2} \int (x^2+2bx+c)^{-\frac{1}{2}} (2x+2b) dx \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \frac{(x^2 + 2bx + c)^{-\frac{1}{2}+1}}{\left(-\frac{1}{2}+1\right)} + c_1 \\
&= \frac{1}{2} \frac{(x^2 + 2bx + c)^{\frac{1}{2}}}{\left(\frac{1}{2}\right)} + c_1 \\
&= \sqrt{x^2 + 2bx + c} + c_1
\end{aligned}$$

Q.7 $\int \frac{\sec^2 x}{\sqrt{\tan x}} dx$

Solution:

$$\begin{aligned}
&\int \frac{\sec^2 x}{\sqrt{\tan x}} dx \\
&= \int (\tan x)^{-\frac{1}{2}} \sec^2 x dx \\
&= \frac{(\tan x)^{-\frac{1}{2}+1}}{\left(-\frac{1}{2}+1\right)} + c \\
&= \frac{(\tan x)^{\frac{1}{2}}}{\left(\frac{1}{2}\right)} + c \\
&= 2(\tan x)^{\frac{1}{2}} \\
&= 2\sqrt{\tan x} + c
\end{aligned}$$

Q.8 (a) Show that

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \ln \left(x + \sqrt{x^2 - a^2} \right) + c$$

Solution:

$$\begin{aligned}
&\int \frac{dx}{\sqrt{x^2 - a^2}} \\
&\text{Let } x = a \sec \theta \Rightarrow \sec \theta = \frac{x}{a} \\
&\text{Then } dx = a \sec \theta \tan \theta d\theta \\
&\therefore \int \frac{dx}{\sqrt{x^2 - a^2}} \\
&= \int \frac{a \sec \theta \tan \theta d\theta}{\sqrt{a^2 \sec^2 \theta - a^2}}
\end{aligned}$$

$$\begin{aligned}
&= \int \frac{a \sec \theta \tan \theta d\theta}{\sqrt{a^2 (\sec^2 \theta - 1)}} \\
&= \frac{a}{a} \int \frac{\sec \theta \tan \theta}{\sqrt{\tan^2 \theta}} d\theta \\
&= \int \sec \theta d\theta \\
&= \ln |\sec \theta + \tan \theta| + c_1 \dots (i)
\end{aligned}$$

$$\begin{aligned}
&\because \sec \theta = \frac{x}{a} \\
&\tan \theta = \sqrt{\sec^2 \theta - 1} \\
&= \sqrt{\frac{x^2}{a^2} - 1} \\
&\Rightarrow \tan \theta = \frac{\sqrt{x^2 - a^2}}{a}
\end{aligned}$$

(i) becomes

$$\begin{aligned}
&= \ln \left| \frac{x}{a} + \frac{\sqrt{x^2 - a^2}}{a} \right| + c_1 \\
&= \ln \left| \frac{x + \sqrt{x^2 - a^2}}{a} \right| + c_1 \\
&= \ln |x + \sqrt{x^2 - a^2}| - \ln a + c_1 \\
&= \ln |x + \sqrt{x^2 - a^2}| + c \quad \because -\ln a + c_1 = c
\end{aligned}$$

(b) Show that

$$\int \sqrt{a^2 - x^2} dx = \frac{a^2}{2} \sin^{-1} \frac{x}{a} + \frac{x}{2} \sqrt{a^2 - x^2} + c$$

Solution:

$$\begin{aligned}
&\int \sqrt{a^2 - x^2} dx \\
&\text{Let } x = a \sin \theta \Rightarrow \theta = \sin^{-1} \frac{x}{a} \\
&\Rightarrow dx = a \cos \theta d\theta \\
&\therefore \int \sqrt{a^2 - x^2} dx \\
&= \int \sqrt{a^2 - a^2 \sin^2 \theta} a \cos \theta d\theta \\
&= \int a \sqrt{1 - \sin^2 \theta} a \cos \theta d\theta \\
&= a^2 \int \sqrt{\cos^2 \theta} \cos \theta d\theta
\end{aligned}$$

$$\begin{aligned}
&= a^2 \int \cos^2 \theta d\theta \\
&= a^2 \int \left(\frac{1 + \cos 2\theta}{2} \right) d\theta \because \cos^2 \theta = \frac{1 + \cos 2\theta}{2} \\
&= \frac{a^2}{2} \left[\int 1 d\theta + \int \cos 2\theta d\theta \right] \\
&= \frac{a^2}{2} \left[\theta + \left(\frac{\sin 2\theta}{2} \right) \right] + c \\
&= \frac{a^2}{2} \left[\theta + \frac{2 \sin \theta \cos \theta}{2} \right] + c \\
&\text{As } x = a \sin \theta \\
&\Rightarrow \sin \theta = \frac{x}{a} \Rightarrow \theta = \sin^{-1} \frac{x}{a} \\
&\Rightarrow \cos \theta = \sqrt{1 - \sin^2 \theta} \\
&\cos \theta = \sqrt{1 - \frac{x^2}{a^2}} = \frac{\sqrt{a^2 - x^2}}{a} \\
&= \frac{a^2}{2} \left[\sin^{-1} \frac{x}{a} + \frac{x}{a} \frac{\sqrt{a^2 - x^2}}{a} \right] + c \\
&= \frac{a^2}{2} \sin^{-1} \frac{x}{a} + \frac{x}{2} \sqrt{a^2 - x^2} + c
\end{aligned}$$

Evaluate the following integrals:

Q.9 $\int \frac{dx}{(1+x^2)^{\frac{3}{2}}}$

Solution:

$$\begin{aligned}
&\int \frac{dx}{(1+x^2)^{\frac{3}{2}}} \\
&\text{Let } x = \tan \theta \\
&\Rightarrow dx = \sec^2 \theta d\theta \\
&\therefore \int \frac{dx}{(1+x^2)^{\frac{3}{2}}} \\
&= \int \frac{\sec^2 \theta d\theta}{(1+\tan^2 \theta)^{\frac{3}{2}}} \\
&= \int \frac{\sec^2 \theta d\theta}{(\sec^2 \theta)^{\frac{3}{2}}}
\end{aligned}$$

$$\begin{aligned}
&= \int \frac{\sec^2 \theta}{\sec^3 \theta} d\theta \\
&= \int \frac{1}{\sec \theta} d\theta \\
&= \int \cos \theta d\theta \\
&= \sin \theta + c \dots (i) \\
&\because \tan \theta = x \Rightarrow \cot \theta = \frac{1}{x} \\
&\sin \theta = \frac{1}{\sqrt{1 + \cot^2 \theta}} = \frac{1}{\sqrt{1 + \frac{1}{x^2}}} = \frac{x}{\sqrt{x^2 + 1}}
\end{aligned}$$

(i) Becomes

$$= \frac{x}{\sqrt{1+x^2}} + c$$

Q.10 $\int \frac{1}{(1+x^2) \tan^{-1} x} dx$

Solution:

$$\begin{aligned}
&\int \frac{1}{(1+x^2) \tan^{-1} x} dx \\
&= \int \frac{1}{\tan^{-1} x} \frac{1}{1+x^2} dx \\
&= \int \frac{1+x^2}{\tan^{-1} x} dx \\
&= \ln |\tan^{-1} x| + c
\end{aligned}$$

Q.11 $\int \sqrt{\frac{1+x}{1-x}} dx$

Solution:

$$\begin{aligned}
&\int \sqrt{\frac{1+x}{1-x}} dx \\
&= \int \sqrt{\frac{1+x}{1-x}} \times \frac{1+x}{1+x} dx \\
&= \int \sqrt{\frac{(1+x)^2}{1-x^2}} dx = \int \frac{1+x}{\sqrt{1-x^2}} dx \\
&= \int \frac{1}{\sqrt{1-x^2}} dx + \int \frac{x}{\sqrt{1-x^2}} dx \\
&= \sin^{-1} x - \frac{1}{2} \int \frac{-2x}{\sqrt{1-x^2}} dx
\end{aligned}$$

$$= \sin^{-1} x - \frac{1}{2} \int (1-x^2)^{-\frac{1}{2}} (-2x) dx$$

$$= \sin^{-1} x - \frac{1}{2} \frac{(1-x^2)^{-\frac{1}{2}+1}}{\left(-\frac{1}{2}+1\right)} + c$$

$$= \sin^{-1} x - \frac{1}{2} \frac{(1-x^2)^{\frac{1}{2}}}{\left(\frac{1}{2}\right)} + c$$

$$= \sin^{-1} x - \sqrt{1-x^2} + c$$

Q.12 $\int \frac{\sin \theta}{1+\cos^2 \theta} d\theta$

Solution:

$$\int \frac{\sin \theta}{1+\cos^2 \theta} d\theta$$

Let $\cos \theta = x \Rightarrow -\sin \theta d\theta = dx$

$$\Rightarrow \sin \theta d\theta = -dx$$

$$\therefore \int \frac{\sin \theta d\theta}{1+\cos^2 \theta}$$

$$= \int \frac{-dx}{1+x^2}$$

$$= -\int \frac{1}{1+x^2} dx$$

$$= -\tan^{-1} x + c$$

$$= -\tan^{-1}(\cos \theta) + c \quad \because \cos \theta = x$$

Q.13 $\int \frac{ax}{\sqrt{a^2-x^4}} dx$

Solution:

$$\int \frac{ax}{\sqrt{a^2-x^4}} dx$$

$$= \int \frac{ax}{\sqrt{a^2-(x^2)^2}} dx$$

Let

$$x^2 = a \sin \theta \Rightarrow \frac{x^2}{a} = \sin \theta \Rightarrow \theta = \sin^{-1} \frac{x^2}{a}$$

$$2x dx = a \cos \theta d\theta$$

$$\Rightarrow x dx = \frac{a}{2} \cos \theta d\theta$$

$$\therefore \int \frac{ax}{\sqrt{a^2-x^4}} dx$$

$$= \int \frac{a \cdot \frac{a}{2} \cos \theta}{\sqrt{a^2-a^2 \sin^2 \theta}} d\theta$$

$$= \frac{a^2}{2} \int \frac{\cos \theta d\theta}{a \sqrt{1-\sin^2 \theta}}$$

$$= \frac{a}{2} \int \frac{\cos \theta d\theta}{\cos \theta}$$

$$= \frac{a}{2} \int d\theta$$

$$= \frac{a}{2} (\theta) + c$$

$$= \frac{a}{2} \sin^{-1} \frac{x^2}{a} + c$$

Q.14 $\int \frac{dx}{\sqrt{7-6x-x^2}}$

Solution:

$$\int \frac{dx}{\sqrt{7-6x-x^2}}$$

$$= \int \frac{dx}{\sqrt{7-6x-x^2-9+9}}$$

$$= \int \frac{dx}{\sqrt{16-(x^2+6x+9)}}$$

$$= \int \frac{dx}{\sqrt{(4)^2-(x+3)^2}}$$

$$\therefore \int \frac{1}{\sqrt{a^2-x^2}} dx = \sin^{-1} \frac{x}{a} + c$$

$$= \sin^{-1} \left(\frac{x+3}{4} \right) + c$$

Q.15 $\int \frac{\cos x}{\sin x \ln \sin x} dx$

Solution:

$$\int \frac{\cos x}{\sin x \ln \sin x} dx$$

Let $\ln \sin x = t$

$$\Rightarrow \frac{1}{\sin x} (\cos x) dx = dt$$

$$\therefore \int \frac{\cos x dx}{\sin x \ln \sin x}$$

$$= \int \frac{dt}{t}$$

$$= \ln |t| + c$$

$$= \ln (\ln \sin x) + c$$

Q.16 $\int \cos x \left(\frac{\ln \sin x}{\sin x} \right) dx$

Solution:

$$\int \cos x \left(\frac{\ln \sin x}{\sin x} \right) dx$$

$$\text{Let } \ln \sin x = t$$

$$\Rightarrow \frac{1}{\sin x} (\cos x) dx = dt$$

$$\therefore \int \cos \left(\frac{\ln \sin x}{\sin x} \right) dx$$

$$= \int t dt$$

$$= \frac{t^2}{2} + c$$

$$= \frac{(\ln \sin x)^2}{2} + c$$

Q.17 $\int \frac{x dx}{4 + 2x + x^2}$

Solution:

$$\int \frac{x dx}{4 + 2x + x^2}$$

$$= \frac{1}{2} \int \frac{2x}{x^2 + 2x + 4} dx$$

$$= \frac{1}{2} \int \frac{2x + 2 - 2}{x^2 + 2x + 4} dx$$

$$= \frac{1}{2} \int \frac{2x + 2}{x^2 + 2x + 4} dx - \frac{1}{2} \int \frac{2}{x^2 + 2x + 4} dx$$

$$= \frac{1}{2} \ln |x^2 + 2x + 4| - \int \frac{1}{x^2 + 2x + 1 + 3} dx$$

$$= \frac{1}{2} \ln |x^2 + 2x + 4| - \int \frac{1}{(x+1)^2 + (\sqrt{3})^2} dx$$

$$= \frac{1}{2} \ln |x^2 + 2x + 4| - \frac{1}{\sqrt{3}} \tan^{-1} \frac{x+1}{\sqrt{3}} + c$$

Q.18 $\int \frac{x}{x^4 + 2x^2 + 5} dx$

Solution:

$$\int \frac{x}{x^4 + 2x^2 + 5} dx$$

$$= \int \frac{x dx}{x^4 + 2x^2 + 1 + 4}$$

$$= \int \frac{x dx}{(x^2 + 1)^2 + (2)^2}$$

$$= \frac{1}{2} \int \frac{2x}{(x^2 + 1)^2 + (2)^2} dx$$

$$= \frac{1}{2} \left(\frac{1}{2} \tan^{-1} \left(\frac{x^2 + 1}{2} \right) \right) + c$$

$$= \frac{1}{4} \tan^{-1} \left(\frac{x^2 + 1}{2} \right) + c$$

Q.19 $\int \cos \left(\sqrt{x} - \frac{x}{2} \right) \times \left(\frac{1}{\sqrt{x}} - 1 \right) dx$

Solution:

$$\int \cos \left(\sqrt{x} - \frac{x}{2} \right) \times \left(\frac{1}{\sqrt{x}} - 1 \right) dx$$

$$\text{Put } \sqrt{x} - \frac{x}{2} = t$$

$$\left(\frac{1}{2\sqrt{x}} - \frac{1}{2} \right) dx = dt$$

$$\frac{1}{2} \left(\frac{1}{\sqrt{x}} - 1 \right) dx = dt$$

$$\left(\frac{1}{\sqrt{x}} - 1 \right) dx = 2dt$$

$$= \int \cos t \cdot 2dt$$

$$\begin{aligned}
 &= 2 \int \cos t \, dt \\
 &= 2 \sin t + c \\
 &= 2 \sin \left(\sqrt{x} - \frac{x}{2} \right) + c
 \end{aligned}$$

Q.20 $\int \frac{x+2}{\sqrt{x+3}} dx$

Solution:

$$\begin{aligned}
 &\int \frac{x+2}{\sqrt{x+3}} dx \\
 &= \int \frac{x+2+1-1}{\sqrt{x+3}} dx \\
 &= \int \frac{x+3-1}{\sqrt{x+3}} dx \\
 &= \int \left(\frac{x+3}{\sqrt{x+3}} - \frac{1}{\sqrt{x+3}} \right) dx \\
 &= \int \sqrt{x+3} dx - \int \frac{1}{\sqrt{x+3}} dx \\
 &= \int (x+3)^{\frac{1}{2}} dx - \int (x+3)^{-\frac{1}{2}} dx \\
 &= \frac{(x+3)^{\frac{3}{2}}}{\left(\frac{3}{2}\right)} - \frac{(x+3)^{\frac{1}{2}}}{\left(\frac{1}{2}\right)} + c \\
 &= \frac{2}{3}(x+3)^{\frac{3}{2}} - 2(x+3)^{\frac{1}{2}} + c
 \end{aligned}$$

Q.21 $\int \frac{\sqrt{2}}{\sin x + \cos x} dx$

Solution:

$$\begin{aligned}
 &\int \frac{\sqrt{2}}{\sin x + \cos x} dx \\
 &= \int \frac{1}{\frac{1}{\sqrt{2}}(\sin x + \cos x)} dx \\
 &= \int \frac{1}{\sin x \frac{1}{\sqrt{2}} + \cos x \frac{1}{\sqrt{2}}} dx \\
 &= \int \frac{1}{\sin x \cos \frac{\pi}{4} + \cos x \sin \frac{\pi}{4}} dx
 \end{aligned}$$

Using $\sin \alpha \cos \beta + \cos \alpha \sin \beta = \sin(\alpha + \beta)$

$$\begin{aligned}
 &= \int \frac{1}{\sin \left(x + \frac{\pi}{4} \right)} dx \\
 &= \int \operatorname{cosec} \left(x + \frac{\pi}{4} \right) dx \\
 &= \ln \left| \operatorname{cosec} \left(x + \frac{\pi}{4} \right) - \cot \left(x + \frac{\pi}{4} \right) \right| + c
 \end{aligned}$$

Q.22 $\int \frac{dx}{\frac{1}{2} \sin x + \frac{\sqrt{3}}{2} \cos x}$

Solution:

$$\begin{aligned}
 &\int \frac{dx}{\frac{1}{2} \sin x + \frac{\sqrt{3}}{2} \cos x} \\
 &= \int \frac{1}{\sin \frac{\pi}{6} \sin x + \cos \frac{\pi}{6} \cos x} \\
 &= \int \frac{dx}{\cos x \cos \frac{\pi}{6} + \sin x \sin \frac{\pi}{6}}
 \end{aligned}$$

Using $\cos \alpha \cos \beta + \sin \alpha \sin \beta = \cos(\alpha - \beta)$

$$\begin{aligned}
 &= \int \frac{dx}{\cos \left(x - \frac{\pi}{6} \right)} \\
 &= \int \sec \left(x - \frac{\pi}{6} \right) dx \\
 &= \ln \left| \sec \left(x - \frac{\pi}{6} \right) + \tan \left(x - \frac{\pi}{6} \right) \right| + c
 \end{aligned}$$